Coefficient Frobenius preserves the full pair

$$\sigma(a+bw) = (a+bw)^7 = a-bw$$

Fixed part
$\sigma(a)=a$
$\mathbb{F}_7$

Negated part
$\sigma(bw)=-bw$
$w\mathbb{F}_7$

Common semilinearity is an extra condition on the same frame and reference.